

Which Value* field is used SHALL be determined based on the data type of the attribute indicated by AttributeID, as described in [the Value* Fields subsection](#).

The AttributeID field SHALL NOT refer to an attribute without the Scenes ("S") designation in the Quality column of the cluster specification.

1.4.7.3.2. ValueUnsigned8, ValueSigned8, ValueUnsigned16, ValueSigned16, ValueUnsigned32, ValueSigned32, ValueUnsigned64, ValueSigned64 Fields

These fields SHALL indicate the attribute value as part of an extension field set, associated with a given AttributeID under an ExtensionFieldSetStruct's ClusterID. Which of the fields is used SHALL be determined by the type of the attribute indicated by AttributeID as follows:

- Data types bool, map8, and uint8 SHALL map to ValueUnsigned8.
- Data types int8 SHALL map to ValueSigned8.
- Data types map16 and uint16 SHALL map to ValueUnsigned16.
- Data types int16 SHALL map to ValueSigned16.
- Data types map32, uint24, and uint32 SHALL map to ValueUnsigned32.
- Data types int24 and int32 SHALL map to ValueSigned32.
- Data types map64, uint40, uint48, uint56 and uint64 SHALL map to ValueUnsigned64.
- Data types int40, int48, int56 and int64 SHALL map to ValueSigned64.
- For derived types, the mapping SHALL be based on the base type. For example, an attribute of type percent SHALL be treated as if it were of type uint8, whereas an attribute of type percent100ths SHALL be treated as if it were of type uint16.
- For boolean nullable attributes, any value that is not 0 or 1 SHALL be considered to have the null value.
- For boolean non-nullable attributes, any value that is not 0 or 1 SHALL be considered to have the value FALSE.
- For non-boolean nullable attributes, any value that is not a valid numeric value for the attribute's type after accounting for range reductions due to being nullable and constraints SHALL be considered to have the null value for the type.
- For non-boolean non-nullable attributes, any value that is not a valid numeric value for the attribute's type after accounting for constraints SHALL be considered to be the valid attribute value that is closest to the provided value.
 - In the event that an invalid provided value is of equal numerical distance to the two closest valid values, the lowest of those values SHALL be considered the closest valid attribute value.

If the used field does not match the data type of the attribute indicated by AttributeID, the AttributeValuePairStruct SHALL be considered invalid.

Examples of processing are:

- ColorControl cluster CurrentX (AttributeID 0x0003) has a type of uint16 and is not nullable.